

Using Machine Learning for the Analysis of Greenland Geodata

Salvador Wahnon Palma (mtr765)

MSc Computer Science, University of Copenhagen

Supervisor: Bulat Ibragimov, Department of Computer Science

Project outside course scope, 15 ECTS, Block 1–2 2026/27

Abstract

Greenland is among the world’s most mineral-rich but least-explored regions. The Geological Survey of Denmark and Greenland (GEUS) publishes decades of exploration data — geochemical samples, scanned exploration reports, known mineral occurrences, intrusions and licence footprints — through open web services. This project investigates what modern machine learning can extract from this heterogeneous archive, in three parts: (0) using a public large language model to parse scanned, text-layer-free exploration reports into structured data; (1) predicting missing element concentrations in the geochemical database with graph neural networks (GNNs) against strong non-graph baselines; and (2) predicting mineral occurrences in areas not covered by local exploration reports. Methodologically, the central finding is that data hygiene dominates model choice: deduplicating the sample database (69,720 rows $\rightarrow$ 14,831 physical samples) and evaluating with spatial block cross-validation change conclusions qualitatively. Under random CV, spatially-aware models appear clearly superior (mean R^2 0.794 vs. 0.748; occurrence ROC-AUC 0.92 vs. 0.91); under leakage-safe spatial CV the advantage largely vanishes (element prediction) or reverses (occurrence prediction, GBDT 0.789 vs. GNN 0.748). The LLM-based report parser achieved exact or near-exact extraction on a synthetic benchmark of scanned-style reports, supporting its use on the full GEUS report archive. We release a reproducible pipeline and a prospectivity map that highlights unexplored candidate areas in East and North-West Greenland.

1 Introduction

Mineral exploration in Greenland has produced a large but fragmented public record. Since the 1950s, companies and surveys have filed exploration reports — typewritten, scanned, and mostly without a machine-readable text layer — while systematic geochemical campaigns have sampled stream sediments and rocks along the ice-free coastal margin. GEUS exposes this archive through public WFS/WMS services and the Greenland Mineral Resources Portal (GEUS): 1,973 georeferenced company reports, almost 100,000 geochemical analyses, 1,274 known mineral occurrences, intrusion outlines, and licence histories.

The data is attractive for machine learning for exactly the reasons it is difficult: it is heterogeneous in provenance, coverage, and quality; labels are one-sided (an occurrence database records where something *was* found, never where nothing exists); and everything is spatially autocorrelated. Following the project agreement with the supervisor, the work is organised as three tasks:

Task 0 — report parsing. Use a public AI model to convert scanned exploration reports into structured records (title, year, commodities, localities, assay values, drilling summaries), and quantify extraction accuracy.

Task 1 — missing geodata. Predict element concentrations that were not measured for a given sample from (a) the sample’s other elements and (b) its spatial neighbourhood, comparing gradient-boosted trees with and without neighbour features against a graph neural network.

Task 2 — occurrence prediction. Predict the presence of mineral occurrences on a 10 km grid, including in cells not covered by any local exploration report, treating the labels as positive-unlabeled.

Contributions. Beyond the three task-specific results, the project makes one methodological point that we believe is the most important outcome for any follow-up work on this archive: *evaluation design, not model architecture, is the first-order term*. The raw geochemistry tables contain each physical sample up to 30 times (once per analytical method); random cross-validation on the raw or even deduplicated data places near-identical or near-adjacent samples on both sides of every split. We quantify the resulting optimism gap for every model, and show that it is large enough to invert model rankings — a GNN that looks like the best occurrence predictor under random CV is the worst under spatial CV. Spatial leakage in geoscientific ML is a known problem (Roberts et al., 2017), but the Greenland archive makes an unusually clean case study because both failure modes (duplication and autocorrelation) are present and separable.

All code, experiments and figures are reproducible from a single Python package (`greenland_ml`), CPU-only, developed against the live GEUS services.¹

2 Data

2.1 Sources

All data comes from the GEUS WFS endpoint (Appendix A). We use four layer groups: *stream-sediment geochemistry* (69,720 rows), *mineral occurrences* (1,274 points with commodity and significance attributes ranging from “showing” to “open mine”), *intrusions* (249 polygons), and *company reports* (1,973 polygons with title, authors, year and commodity metadata). Rock-sample geochemistry (26,068 rows) is available but geographically restricted to West/South Greenland (lat. 59.9–70.2°N); we use stream sediments throughout as they cover the whole ice-free margin (Fig. 1).

2.2 Duplication and cleaning

The raw stream table has one row per (physical sample, analytical method): the same field sample analysed by, e.g., XRF and INAA appears as several rows, each with a different subset of elements filled in — on average 4.5 rows per sample, up to 30. We merge to one row per `sample_number`, taking per element the first above-detection value and otherwise the first reported (below-detection) value, then drop samples with fewer than three above-detection measurements or missing coordinates. This yields **14,831 physical samples**. Negative values encode “below detection limit” and are treated as missing; concentrations are $\log_{10}$ -transformed for all modelling.

We keep the 15 elements measured in at least 35% of raw rows; after merging, their per-sample coverage is 80–98% (CaO, Zn, TiO₂, K₂O, MnO, Rb, Fe₂O₃, Na₂O, Cu, Cr, Ba, V, Sr, U, Co), giving a $14,831 \times 15$ matrix with 88.6% of entries observed.

¹Data endpoints listed in Appendix A. The stream-sediment dataset is also archived at GEUS Dataverse (GEUS, 2022).

2.3 Spatial evaluation protocol

Stream-sediment geochemistry is strongly spatially autocorrelated — neighbouring catchments drain the same lithologies (Tobler, 1970). We therefore evaluate everything with **spatial block cross-validation** (Roberts et al., 2017): points are binned into 50 km squares (100 km for Task 2’s coarser cell grid), and whole blocks are randomly assigned to five folds, so test samples are never a few kilometres from training samples. As an explicit measurement of leakage — not as a result — every experiment is repeated with plain random five-fold CV.

3 Task 0: Parsing scanned exploration reports with an LLM

3.1 Problem

The 1,973 archived reports carry structured metadata in the WFS layer, but their content — assay tables, drilling logs, field observations, recommendations — exists only as scanned typewriter pages. `pypdf` extracts zero characters from the PDFs we received: there is no text layer, so classical scraping is impossible and OCR must be visual. The hypothesis of Task 0 is that a current vision-capable LLM can read these scans directly and emit faithful structured records, turning the whole archive into a queryable database and, eventually, into node attributes for the graphs of Tasks 1–2.

3.2 Pipeline

We define a fixed JSON schema per report: file number, title, authors, company, year, commodities, named localities with decimal-degree coordinates, work performed, assay results (sample id, element, value, unit, context), drilling summary (holes, total metres), conclusions and recommendations, plus free-text extraction notes. A batch driver invokes the Claude Code CLI (Anthropic) headlessly — one call per PDF — on a Claude subscription; the agent reads the scanned pages visually (in 20-page chunks, prioritising title page, summary, conclusions and assay tables for long reports), and must return a single schema-conformant JSON object. Returned objects are validated (types, required keys, plausible years, coordinates inside Greenland’s bounding box, numeric assay values); failures and warnings are logged, and the run is resumable. The extraction prompt forbids guessing: absent or illegible fields must be null, with the uncertainty noted.

Two implementation details mattered in practice. First, prompts are piped via `stdin`: the Windows CLI shim truncates multi-line prompt arguments at the first newline, which silently destroys the instruction set. Second, the agent requires a local PDF rasteriser (poppler) — the pipeline documents both dependencies.

3.3 Evaluation design

Extraction accuracy needs ground truth. We use two complementary sources:

(a) Synthetic scanned reports. We generated three period-plausible exploration reports (copper–zinc exploration with drilling, a uranium stream-sediment reconnaissance, a graphite occurrence note) with known ground truth, typeset in a typewriter font, *rasterised* page by page with random small rotations and sensor-like noise, and embedded as images — like the real scans, the resulting PDFs contain no text layer (`pdftotext` extracts nothing). This benchmark tests the full pipeline end to end with exact labels, including degrees-minutes to decimal coordinate conversion and tabular assay extraction.

Table 1: Task 0 extraction accuracy on the synthetic scanned-report benchmark (three reports, exact ground truth). Assay recall/precision match (element, value) pairs after unit-preserving canonicalisation.

Report	Title sim.	Year	Authors	Commodities (J)	Assay rec.	Assay prec.	Drilling
90001 (Cu–Zn)	1.00	✓	1.00	1.00	1.00	1.00	✓
90002 (U)	1.00	✓	1.00	1.00	1.00	1.00	✓
90003 (graphite)	1.00	✓	1.00	1.00	1.00	1.00	✓

(b) WFS metadata cross-reference. For the real PDFs, the WFS report layer provides independently curated title, year, authors and commodities. Agreement between extraction and WFS metadata (title similarity, year exactness, commodity-set Jaccard after canonicalisation) estimates real-document accuracy on those fields without manual labelling; body fields (assays, drilling) still require spot checks.

3.4 Results

On the synthetic benchmark, all three reports were extracted successfully (schema-valid on first attempt, ~ 20 s per report with Claude Sonnet). Field-level accuracy is shown in Table 1. After canonical element naming (“graphitic carbon” $\equiv$ C/graphite), every extracted title, year, author list, commodity set, assay value and drilling count matched ground truth exactly; degrees-minutes coordinates were correctly converted to (negative-longitude) decimal degrees. The model also translated Danish-style phrasing and populated `extraction_notes` sensibly (e.g. flagging that two samples were reported lost and not assayed).

[TODO: Run the pipeline on the 20 real GEUS PDFs (copy them to `data/reports_pdf/` as `<file_no>.pdf`) and report the WFS cross-reference metrics here, plus a manual spot check of assay tables. The synthetic benchmark bounds pipeline correctness, not real-scan difficulty: genuine 1950s scans have skew, bleed-through and stamp overprints that the synthetic pages only approximate.]

3.5 Limitations

The synthetic scans are cleaner than the worst archive pages, and three documents is a pipeline test, not a corpus-level accuracy estimate. Costs scale linearly in pages; extraction of all 1,973 reports is feasible but should batch by report length and use the WFS cross-reference as an automatic acceptance test.

4 Task 1: Predicting missing geochemistry

4.1 Problem

Given a sample with some elements measured, predict a held-out element’s $\log_{10}$ concentration. This is the imputation problem that would fill the 11.4% missing entries of the element matrix, and it doubles as a controlled test of whether *spatial context* carries information beyond a sample’s own chemistry — the question that decides whether graph models are worth pursuing on this archive.

4.2 Models

All models are evaluated per target element e on identical folds and test samples (those with e observed).

Table 2: Task 1: pooled test R^2 per element ($\log_{10}$ scale), spatial block CV (5 folds, 50 km blocks). Best per row in bold. Elements sorted by GBDT R^2 . The median baseline’s pooled R^2 is ≈ 0 by construction and slightly negative under block CV because fold distributions shift.

Element	Median	GBDT	GBDT+NN	GNN
Fe ₂ O ₃	−0.018	0.854	0.849	0.819
Na ₂ O	−0.106	0.822	0.823	0.769
Co	−0.009	0.782	0.756	0.753
MnO	−0.003	0.778	0.766	0.734
K ₂ O	−0.019	0.774	0.762	0.732
TiO ₂	−0.004	0.764	0.778	0.727
Rb	−0.005	0.730	0.725	0.697
Cu	−0.016	0.697	0.700	0.650
Sr	−0.024	0.655	0.679	0.596
CaO	−0.015	0.651	0.621	0.623
Ba	−0.013	0.627	0.627	0.573
Zn	−0.019	0.591	0.577	0.561
Cr	−0.006	0.590	0.589	0.569
V	−0.018	0.532	0.533	0.503
U	−0.036	0.472	0.578	0.431
mean	−0.021	0.688	0.691	0.649

Median. Training-fold median of e (sanity floor).

GBDT. A histogram gradient-boosted regressor (Pedregosa et al., 2011) on the sample’s other 14 elements (missing values handled natively; 300 iterations, learning rate 0.08).

GBDT + neighbours. The same, plus 16 spatial-context features: per-element medians over the $k=8$ nearest *training* samples (self-matches excluded) and the log mean neighbour distance. Neighbour features are computed strictly from training folds, so they encode legitimate historical context rather than test information.

GNN. A masked-feature imputation network on the sample kNN graph ($k=8$; median edge length 4.1 km): two GraphSAGE layers (Hamilton et al., 2017) (hidden 128) with an input-skip connection into a two-layer head that predicts all 15 standardised element values. Node input is [values with missing/masked entries zeroed || observed mask] (30 channels). Training is a denoising objective: each epoch a random 30% of observed entries are hidden and reconstructed (400 epochs, Adam, lr $5 \cdot 10^{-3}$). Leakage control is *inductive*: the model trains only on the subgraph induced by training-fold nodes; at evaluation the full graph is used with the target element zeroed on *all* test nodes simultaneously, so no test value of the target is visible anywhere in the input. Implementation in PyTorch Geometric (Fey and Lenssen, 2019).

4.3 Results

Three observations (Table 2, Figs. 2–3):

(1) **Chemistry alone is a strong predictor.** Plain GBDT on co-measured elements reaches mean R^2 0.688 under spatial CV — major-oxide chemistry (Fe₂O₃, Na₂O, MnO, K₂O) is the most predictable, trace metals (Zn, Cr, V, U) the least. This confirms, now under leakage-safe evaluation, the strong element co-prediction observed in the supervisor’s preliminary random-CV experiments.

(2) **Spatial context adds almost nothing on average — once leakage is removed.** Under random CV, neighbour features add +0.047 mean R^2 and the GNN beats plain GBDT;

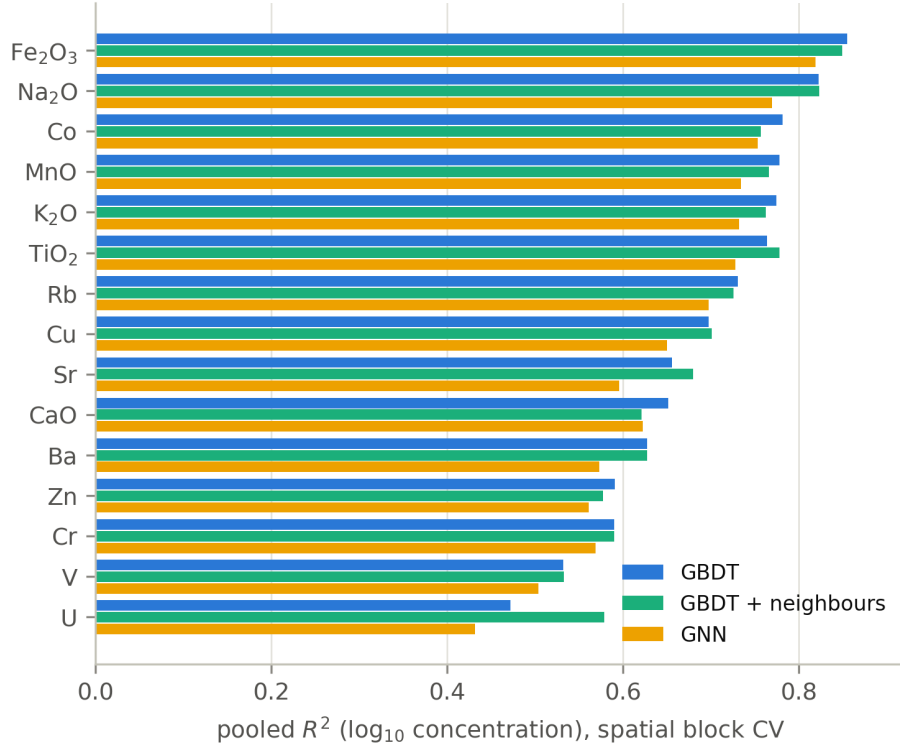


Figure 2: Task 1 per-element pooled R^2 under spatial block CV.

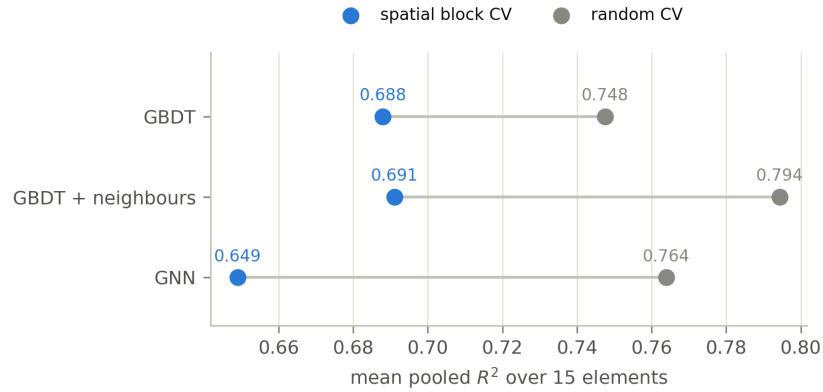


Figure 3: The leakage gap. Mean pooled R^2 of each model under random CV (grey) versus spatial block CV (blue). The spatially-aware models lose the most: their random-CV advantage is largely reproduced test-set information from nearby samples.

under spatial block CV the neighbour advantage collapses to +0.003 and the GNN trails GBDT by 0.039. The apparent value of spatial context was mostly the model reading the test set through its neighbours.

(3) The exception is informative. Uranium is the one element where neighbours help substantially even under spatial CV ($0.472 \rightarrow 0.578$, +0.106). This is geologically plausible: uranium anomalies in South Greenland form spatially coherent districts (e.g. around the Ilímaussaq complex) that extend beyond single catchments, so a 50 km block held out for testing still borders genuinely informative training regions. Spatial models earn their keep exactly where mineralisation is district-scale — which is encouraging for Task 2, whose target is precisely district-scale structure.

4.4 Why doesn’t the GNN win?

Under honest evaluation the GNN is a competitive imputer (mean 0.649) but not better than trees. We see three non-exclusive explanations, which map directly to future work: (i) with only element concentrations as node features, messages passed between neighbours carry little that the neighbour-median features don’t already summarise, and gradient boosting is simply a stronger tabular learner (Grinsztajn et al., 2022); (ii) our inductive masking protocol is deliberately conservative — test nodes contribute no features during training — trading accuracy for a clean leakage guarantee; (iii) the graph is purely geometric (kNN), while the geologically meaningful adjacency (shared catchment, shared lithology, same intrusive suite) is not yet encoded. Enriching nodes and edges with Task-0 report attributes and geological map units is the natural next step.

5 Task 2: Predicting occurrences beyond explored areas

5.1 Setup

We discretise the sampled margin into a 10 km grid, keeping the 7,280 cells within 30 km of at least one stream sample. Cell features (18): per-element medians of the 10 nearest samples, log mean distance to those samples, log sample density within 10 km, and log distance to the nearest mapped intrusion. A cell is labelled positive if a known occurrence lies within 10 km of its centre — 1,167 cells (16.0%).

Positive-unlabeled labels. An unlabelled cell is not a negative: it may simply never have been visited. We quantify this with an *exploration intensity* covariate — the number of company-report polygons with local footprints ($\leq 10^4$ km² bounding box; 1,422 reports) covering the cell. 31.5% of cells are locally explored, and the occurrence rate is $5.7\times$ higher there (36.8% vs. 6.5%): the label process is strongly observation-biased, the defining condition of PU learning (Elkan and Noto, 2008). We adopt the standard non-traditional classifier: unlabelled cells are treated as negatives with positives re-weighted (balanced class weights / BCE pos-weight), and scores are read as relative rankings rather than calibrated probabilities.

Models. GBDT classifier (balanced class weights) versus a two-layer GraphSAGE node classifier with input skip (hidden 64, BCE with positive re-weighting, 300 epochs) on the cell kNN graph ($k=8$, effectively grid adjacency). Evaluation: spatial block CV with 100 km blocks (plus the random-CV leakage rerun), reporting ROC-AUC and average precision against the PU labels — both are then *lower bounds* on performance against the unknown true labels.

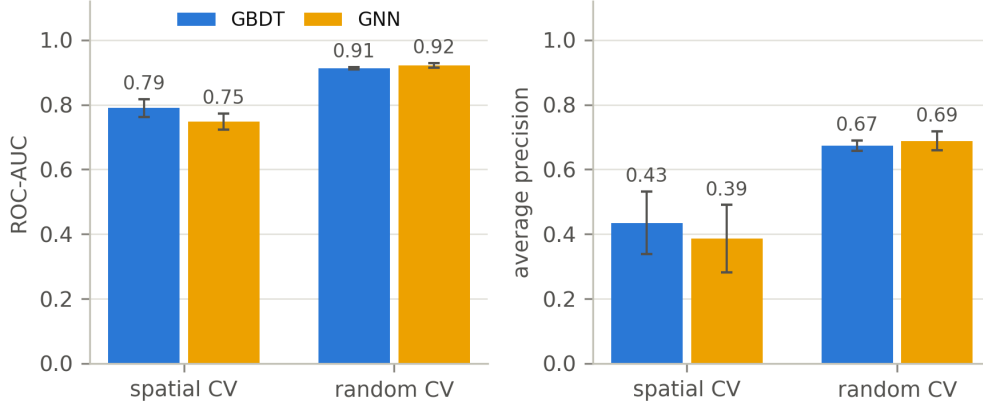


Figure 4: Task 2 cross-validated performance (mean $\pm$ std over 5 folds). Random CV not only inflates both models by ~ 0.15 ROC-AUC but *reverses their ranking*.

Table 3: Highest-scoring unexplored, occurrence-free cells (GNN score; GBDT score for comparison). Coordinates are cell centres (WGS84).

Lon	Lat	GNN	GBDT	District
−26.53	73.58	0.99	0.35	central East Greenland
−71.25	77.97	0.99	0.18	Melville Bugt / Thule
−65.91	77.66	0.99	0.48	Melville Bugt / Thule
−69.62	76.52	0.99	0.87	Melville Bugt / Thule
−72.89	77.98	0.98	0.28	Melville Bugt / Thule
−53.50	68.48	0.97	0.57	northern Disko Bugt
−58.43	82.20	0.96	0.04	North Greenland

5.2 Results

Under spatial CV, GBDT reaches ROC-AUC 0.789 ± 0.028 and average precision 0.435 ± 0.097 (vs. a 16% positive base rate); the GNN trails at 0.748 ± 0.025 / 0.386 ± 0.104 (Fig. 4). Under random CV both models exceed 0.91 ROC-AUC and the GNN nominally *wins* — a ranking inversion that would have led to exactly the wrong architectural conclusion had we not used spatial evaluation. Discrimination well above chance at 100 km block separation indicates the models capture genuine regional geochemical signatures of mineralisation, not just proximity to known finds.

Fig. 5 shows the resulting prospectivity surface. Among *unexplored, occurrence-free* cells, the highest-scoring candidates (Table 3) cluster in three districts: central East Greenland around 73–74°N (adjacent to the known Mesters Vig–Ymer Ø Pb–Zn province), the Melville Bugt/Thule margin in the North-West (76–78°N), and the northern Disko Bugt region. These are plausible follow-up targets precisely because the model ranks them like explored mineralised districts while no local report covers them.

5.3 Caveats

PU metrics understate true performance but also hide observation bias: a model can score well partly by recognising “places geologists visit”. Our density and sample-distance features let models explain some acquisition effects away, and the intensity covariate quantifies the bias, but a principled treatment (e.g. occurrence-significance weighting, bias-corrected PU risk estimators, or holding out entire *explored* districts as pseudo-unexplored tests) is future work. The candidate list should be read as “geochemically analogous to mineralised districts”, not as

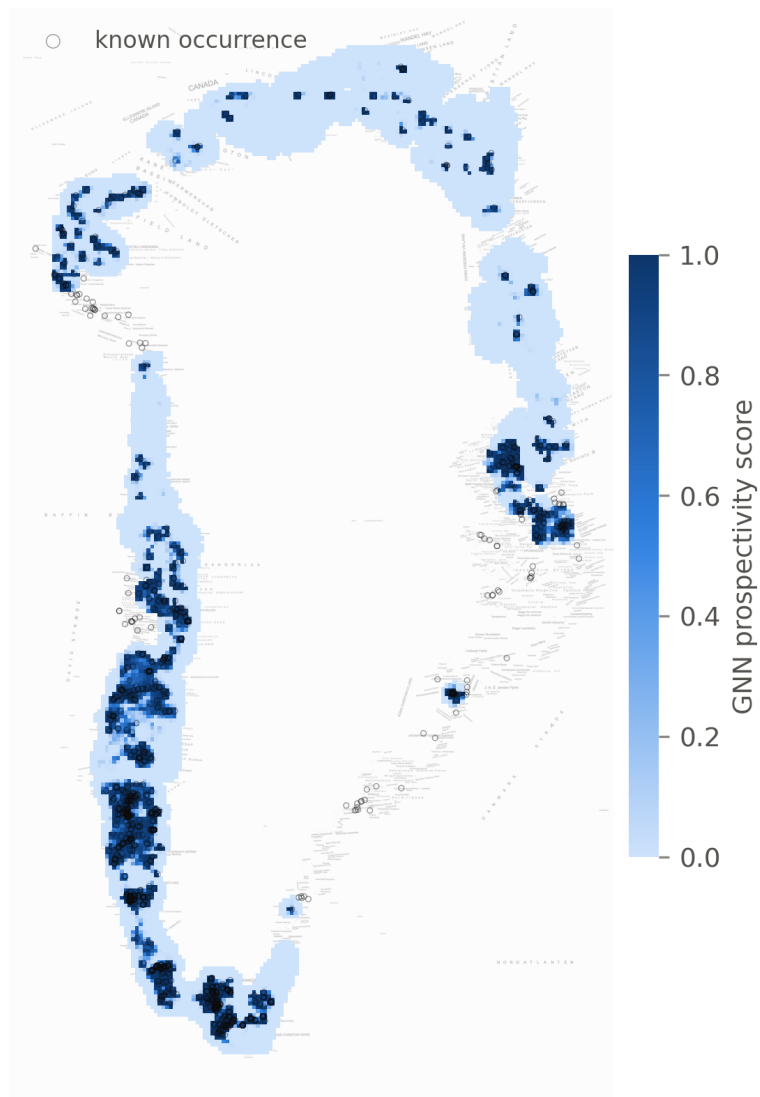


Figure 5: GNN prospectivity scores (final model trained on all cells) with known occurrences overlaid (black circles). High scores concentrate in South Greenland, the Disko–Nuussuaq region and central East Greenland; the model also scores several occurrence-free, unexplored districts highly (Table 3).

drill targets.

6 Discussion

What we learned about this archive. (1) The single most consequential modelling step was not a model: merging method-level duplicate rows and blocking CV spatially. Together they turn apparent effects (spatial features $+0.05 R^2$; GNN best-in-class occurrence AUC 0.92) into measured leakage. Any future work on GEUS data that reports random-CV numbers should be read with this gap in mind. (2) Sample chemistry is highly self-redundant (mean R^2 0.69 from other elements alone), so imputation of the archive’s missing entries is already practical with trees. (3) Spatial/graph structure pays off selectively — where mineral systems are district-scale (U in South Greenland; regional occurrence prediction at AUC 0.79 across 100 km blocks) — and the way to make GNNs win is to give them geologically meaningful

graphs and richer node attributes, not more epochs.

Why the three tasks form one project. Task 0 turns unstructured reports into attributes; Tasks 1–2 currently run on geometry and chemistry alone. The obvious composition — report-derived node features (commodities sought, work done, assay highlights) attached to samples and cells, and report–area edges forming a heterogeneous graph — is exactly the setting where message passing has information that tabular baselines cannot see. That composition is the natural thesis-scale extension.

Limitations. Single dataset (stream sediments); GNN hyperparameters tuned only lightly (no architecture search); coordinate conversion uses a series approximation of UTM 24N (metre-level inside the zone, up to tens of metres at Greenland’s east/west extremes — negligible at 10–50 km scales); the Task 2 cell graph is transductive in features (labels of test cells are never used); occurrence positives ignore significance grading (a “showing” counts like a mine); and the Task 0 corpus-level accuracy on real scans is still to be measured (Sec. 3).

7 Conclusion and future work

We built and evaluated a complete, reproducible ML pipeline over the public GEUS Greenland archive, organised around three tasks: LLM-based parsing of scanned exploration reports (validated end-to-end on a synthetic scanned benchmark with exact-match extraction), missing-geochemistry prediction (trees at mean R^2 0.69 under spatial CV; GNN competitive but not superior), and occurrence prospectivity beyond locally explored areas (ROC-AUC 0.79 at 100 km block separation, with a candidate map of unexplored districts). The recurring methodological result is that deduplication and spatial block CV are prerequisites for any valid conclusion on this archive: they shrink or invert every “spatial model wins” effect measured under random CV.

Future work, in order of expected value: (i) run Task 0 over the full 1,973-report corpus with WFS cross-reference acceptance tests; (ii) build the heterogeneous graph (samples, cells, reports, intrusions, occurrences) with report-derived attributes and geological map units as node/edge features, and re-pose Tasks 1–2 on it; (iii) add geophysical rasters (aeromagnetics) as cell features; (iv) bias-aware PU objectives and significance-weighted labels for Task 2; (v) uncertainty quantification (block bootstrap, conformal prediction) so the prospectivity map reports confidence, not just score.

Declaration on the use of AI tools

This project used Claude (Anthropic), via the Claude Code environment, in two roles, both with my supervisor’s knowledge and encouragement: (1) as a programming assistant for the `greenland_ml` package, experiments and figures — all reported numbers are produced by running the included code against the public GEUS services and are reproducible with the commands in the repository README; and (2) as the extraction engine *studied* in Task 0, where the model is the object of evaluation. AI assistance was also used in drafting this report from the experimental outputs; I have reviewed, verified and revised the content and take full responsibility for it. [TODO: Salvador: revise the report in your own voice, then align this declaration’s wording with the current UCPH/SCIENCE rules on AI-assisted work before submission.]

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A Data sources and reproduction

Table 4: GEUS WFS layers used (endpoint `data.geus.dk/geusmap/ows/32624.jsp`, EPSG:32624; counts verified 2026). GetFeature must use WFS 1.1.0/GML — the JSON output format and WFS 2.0.0 GetFeature fail on several layers.

Layer	Count	Use
<code>ms:geochemistry_stream_raw</code>	69,720	Tasks 1–2 features
<code>ms:mineral_occurrences_v3_external</code>	1,274	Task 2 labels
<code>ms:dodex_company_reports</code>	1,973	Task 0 reference; Task 2 exploration intensity
<code>ms:intrusions</code>	249	Task 2 features
<code>ms:geochemistry_rock_samples_open</code>	26,068	held out (regional coverage only)

Reproduction: `pip install -r requirements.txt`, then `python -m greenland_ml.task1_missing.experiment` (~20 min CPU), `python -m greenland_ml.task2_occurrences.experiment` (~4 min), `python -m greenland_ml.viz.figures`. Task 0 additionally requires the Claude Code CLI and poppler on `PATH`; see README. Fixed seeds throughout (42).